

From Photons to Features:
Physics, Signal Processing, and Invariant State-Space
Architectures
in 77 GHz Automotive Radar Perception

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Part I

Electromagnetic Foundations & FMCW Front-End Physics

Chapter 1

The Monostatic Radar Paradigm & FMCW Principles

1.1 Intuition: echoes, pitch, and the laser tape measure

Stand in a canyon and shout. A fraction of a second later the shout comes back. Children learn, without any mathematics, the two great facts of echolocation: (i) the *delay* of the echo tells you how *far* the wall is, and (ii) if you shout from a moving car, the returning echo sounds *shifted in pitch*. Bats turned these facts into a hunting sensor a hundred million years before engineers did; a bat's chirp is a frequency-modulated continuous wave, and this book is, in a precise sense, the story of how the modern automobile learned the bat's trick at 77 GHz.

A laser tape measure does the same thing with light: it times a pulse. Automotive radar cannot afford the peak power of short pulses, nor the blanking interval during which a pulsed receiver is deaf. Instead it *chirps continuously*: the transmitted frequency ramps linearly upward, like a siren sliding from low to high, millions of times per second. The echo returns a slightly older copy of that slide. Mixing (multiplying) the fresh transmit slide with the stale echo produces a low, audible-like tone — the *beat frequency* — whose pitch is distance and whose slow drift from chirp to chirp is velocity. Everything in this chapter is the quantitative unfolding of that paragraph.

1.2 The 77 GHz automotive band

Regulators allocate 76–81 GHz to automotive radar. The carrier wavelength is

$$\lambda = \frac{c}{f_c} = \frac{2.99792458 \times 10^8}{77 \times 10^9} \approx 3.896 \times 10^{-3} \text{ m} \approx 3.896 \text{ mm}. \quad (1.1)$$

Three consequences follow. First, antennas shrink to millimetres, so a dozen transmitters and receivers fit behind a bumper emblem. Second, a 3.9 mm wave treats guardrails, bridge decks, and car bodies as *smooth mirrors* (roughness $\ll \lambda$), which is why specular multipath ghosts — the subject of chapter 12 — dominate automotive scenes. Third, rain backscatter at 77 GHz is strong enough to blind naive detectors (tire spray), yet the R^{-4} two-way path loss still lets a 10 dBsm car be seen past 150 m; optical sensors, by contrast, die completely in fog that radar barely notices. The one-way Friis power at range r falls as r^{-2} ; the echo must come home, so radar pays r^{-4} :

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 r^4}, \quad (1.2)$$

with transmit power P_t , antenna gain G , and radar cross section σ . Doubling the range costs 12 dB of signal.

1.3 Why continuous wave and not pulses

A pulsed radar shouts and then cups its ear: during the microsecond shout it hears nothing (eclipse), and the shout must be kilowatts to survive the R^{-4} toll. A car bumper offers milliwatts and demands minimum ranges of a few metres (parking, cut-in). Linear frequency modulated continuous wave (LFMCW) transmits *while* receiving, at milliwatts, with no minimum range beyond one ramp. The price is self-interference (the transmitter shouts directly into its own ear), paid with careful RF isolation and the fact that the beat signal of interest sits far above the leakage pedestal.

1.4 The LFMCW ramp, line by line

Let the chirp bandwidth be B (Hz), the sweep duration T_{chirp} (s), and the chirp slope

$$S = \frac{B}{T_{\text{chirp}}} \quad [\text{Hz/s}]. \quad (1.3)$$

The transmitted waveform over $0 \leq t < T_{\text{chirp}}$ is

$$s_{\text{tx}}(t) = A_{\text{tx}} \cos(2\pi f_c t + \pi S t^2 + \phi_0). \quad (1.4)$$

Its instantaneous frequency is the phase derivative divided by 2π :

$$f_{\text{inst}}(t) = \frac{1}{2\pi} \frac{d}{dt} (2\pi f_c t + \pi S t^2) = f_c + S t, \quad (1.5)$$

a straight ramp from f_c to $f_c + B$. Suppose a point target sits at range r_0 with radial velocity v_r (positive receding). The round-trip delay is

$$\tau(t) = \frac{2(r_0 + v_r t)}{c}. \quad (1.6)$$

The factor of 2 is mandatory and is the single most common beginner trap: the wave must travel *to* the target and *back*. Halving it produces the “12.5 MHz one-way trap” of section 1.5. The echo is the transmit waveform evaluated at retarded time:

$$s_{\text{rx}}(t) = A_{\text{rx}} \cos(2\pi f_c(t - \tau) + \pi S(t - \tau)^2 + \phi_0). \quad (1.7)$$

1.5 Heterodyne downmixing: the beat is born

The mixer multiplies the two waveforms. With Φ_{tx} , Φ_{rx} the two cosine arguments, the product-to-sum identity

$$\cos \Phi_{\text{tx}} \cos \Phi_{\text{rx}} = \frac{1}{2} \cos(\Phi_{\text{tx}} - \Phi_{\text{rx}}) + \frac{1}{2} \cos(\Phi_{\text{tx}} + \Phi_{\text{rx}}) \quad (1.8)$$

splits the product into a *difference* term (slow, kilohertz to megahertz — our signal) and a *sum* term near $2f_c \approx 154$ GHz, which the low-pass filter annihilates. The surviving phase difference is

$$\begin{aligned} \Delta\Phi(t) &= [2\pi f_c t + \pi S t^2] - [2\pi f_c(t - \tau) + \pi S(t - \tau)^2] \\ &= 2\pi f_c \tau + \pi S(2t\tau - \tau^2). \end{aligned} \quad (1.9)$$

Differentiating and dividing by 2π gives the instantaneous beat frequency

$$f_b(t) = f_c \dot{\tau} + S\tau + St\dot{\tau} - S\tau\dot{\tau}. \quad (1.10)$$

Now $\dot{\tau} = 2v_r/c \sim 10^{-7}$ for highway speeds, so every term carrying $\dot{\tau}$ except the first is parts-per-million next to $S\tau$. Dropping them (we will resurrect the Doppler term properly in slow time in chapter 2),

$$f_b = S\tau = \frac{B}{T_{\text{chirp}}} \cdot \frac{2r}{c} = \frac{2Br}{cT_{\text{chirp}}}. \quad (1.11)$$

Worked example. $B = 1 \text{ GHz}$, $T_{\text{chirp}} = 40 \mu\text{s}$, $r = 150 \text{ m}$:

$$f_b = \frac{2 \times 10^9 \times 150}{3 \times 10^8 \times 40 \times 10^{-6}} = \frac{3 \times 10^{11}}{1.2 \times 10^4} = 25 \text{ MHz}. \quad (1.12)$$

Forgetting the round trip gives 12.5 MHz — the one-way trap. Every range bin in every radar on the road is eq. (1.11) in disguise.

1.6 Quadrature demodulation: why one mixer is not enough

A single real mixer outputs $\cos(2\pi f_b t)$: but $\cos(\omega t) = \cos(-\omega t)$, so approaching and receding targets at the same $|f_b|$ are indistinguishable (sign ambiguity), and any fading null kills the signal. Splitting the local oscillator into 0° and 90° branches yields the in-phase and quadrature channels

$$I(t) = A \cos(2\pi f_b t + \phi_0), \quad Q(t) = A \sin(2\pi f_b t + \phi_0), \quad (1.13)$$

combined into the complex analytic signal

$$x(t) = I(t) + jQ(t) = A e^{j(2\pi f_b t + \phi_0)}. \quad (1.14)$$

Negative frequencies now rotate clockwise, positive counter-clockwise: the sign ambiguity is gone, and $|x(t)| = A$ never fades. The dual ADC samples eq. (1.14); everything downstream is complex arithmetic on vector DSPs.

1.7 Figures: transceiver and chirp geometry

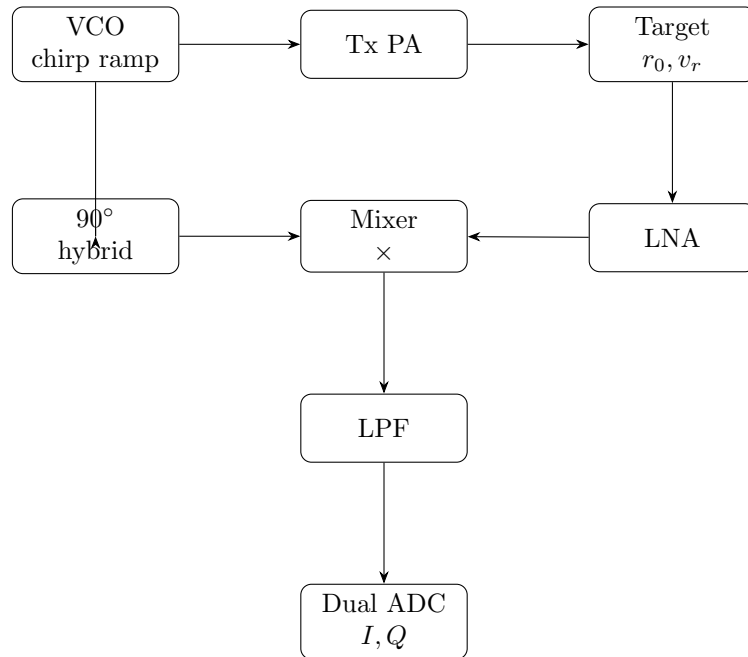


Figure 1.1: RF transceiver: the VCO chirp feeds the power amplifier and, as LO, the quadrature mixer; the echo returns through the LNA. The 154 GHz sum term dies in the LPF.

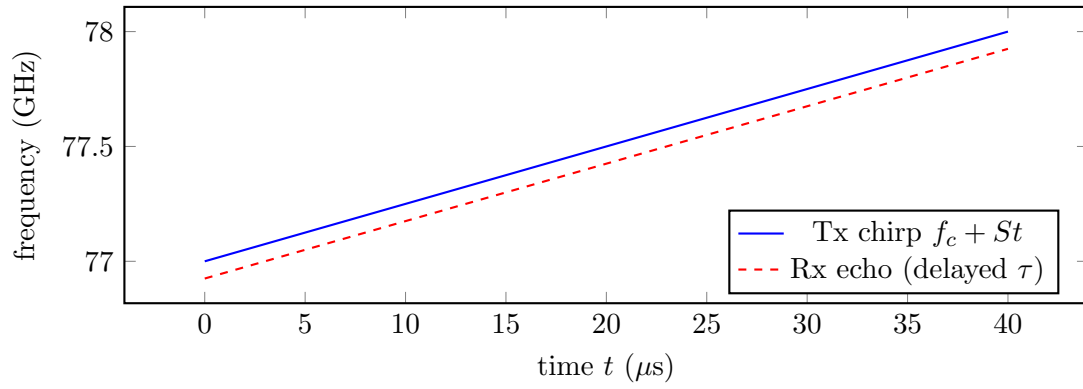


Figure 1.2: Frequency versus time: the echo is a delayed copy of the ramp; the vertical gap is the beat frequency $f_b = S\tau$ of eq. (1.11).

Part II

Two-Dimensional Spectral Estimation & Data Cube Architecture

Chapter 2

The Fast-Time / Slow-Time Two-Dimensional DSP Chain

2.1 Intuition: ripples in a pond, frame by frame

Throw a stone into a pond and photograph the ripples. One photograph tells you the wavelength (distance between crests); a *movie* of photographs tells you how fast the pattern marches outward. Range-Doppler processing is exactly this: each chirp is one photograph (fast time gives range from the beat pitch), and the stack of chirps is the movie (slow time gives velocity from how the phase marches from frame to frame). The mathematics below is only the quantitative version of “compare consecutive frames.”

2.2 Fast time: sampling the beat, the Range-FFT

The dual ADC samples eq. (1.14) at rate f_s (interval $T_s = 1/f_s$), collecting N_R fast-time samples per chirp from chirp m : $x_m[n]$, $n = 0, \dots, N_R - 1$. The windowed discrete Fourier transform is

$$X_m[k] = \sum_{n=0}^{N_R-1} x_m[n] w_R[n] e^{-j \frac{2\pi}{N_R} nk}, \quad k = 0, \dots, N_R - 1. \quad (2.1)$$

A pure beat tone of frequency f_b peaks at bin $k^* = f_b N_R T_s$. Substituting eq. (1.11),

$$r_m = k \cdot \frac{c}{2ST_s N_R} = k \cdot \text{rbin_res}, \quad (2.2)$$

so each bin is a tape-measure tick of width `rbin_res` (a Q16.16 fixed-point field, `Look_Data.T.rbin_res`). The *resolution* — the closest two targets that do not merge — comes from the DFT Rayleigh limit $\Delta f = 1/(N_R T_s)$ mapped through eq. (1.11):

$$\boxed{\Delta r = \frac{c}{2B}}. \quad (2.3)$$

Only bandwidth buys resolution: $B = 1\text{ GHz}$ gives 15 cm, no matter how long the chirp. Windowing (w_R , Hann/Hamming) trades 1.5–2× mainlobe widening for sidelobe suppression; the window’s coherent loss is calibrated out in `range_proc.c`.

2.3 Slow time: the movie between chirps

Transmit N_D chirps spaced by the repetition time T_c . During one chirp the target barely moves, but between chirps it walks $\Delta d = v_r T_c$. The round trip grows by $2\Delta d$, i.e. a phase advance per

chirp of

$$\Delta\Phi = \frac{2\pi}{\lambda} 2v_r T_c = \frac{4\pi v_r T_c}{\lambda}. \quad (2.4)$$

The factor 4π (not 2π) is the round trip striking again: the phase ruler is half-wavelengths. At fixed range bin k , the slow-time sequence $X_m[k]$, $m = 0, \dots, N_D - 1$, is a sampled sinusoid of “frequency” $\Delta\Phi$; its DFT (Doppler-FFT)

$$Y[k, l] = \sum_{m=0}^{N_D-1} X_m[k] w_D[m] e^{-j \frac{2\pi}{N_D} ml} \quad (2.5)$$

peaks at $l^* = N_D \Delta\Phi / 2\pi$, giving

$$v_l = l \cdot \frac{\lambda}{2N_D T_c} = l \cdot \text{dbin_res}, \quad v_{\max} = \frac{\lambda}{4T_c}. \quad (2.6)$$

The Nyquist limit bites at $\Delta\Phi = \pm\pi$: faster targets wrap around the Doppler spectrum (aliasing), which Gen8 unfolds with staggered PRFs (chapter 6). With $\lambda = 3.896$ mm and $T_c = 40 \mu\text{s}$, $v_{\max} \approx 24.4$ m/s per sequence.

2.4 The Range-Doppler matrix

$|Y[k, l]|^2$ is the Range-Doppler power map: columns are range bins, rows are Doppler bins, bright cells are targets. Every later stage — detection (chapter 4), beamforming (chapter 5), rendering (section 14.2) — reads this matrix.

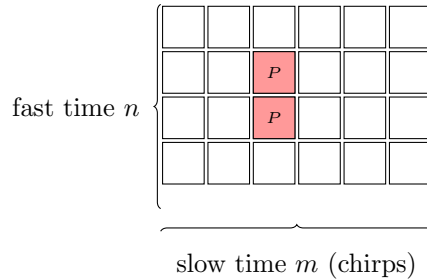


Figure 2.1: Sampling grid: fast-time ADC samples run down each chirp column; the Doppler-FFT runs across the slow-time row. The red cells are one target’s range bin marching in phase down the column.

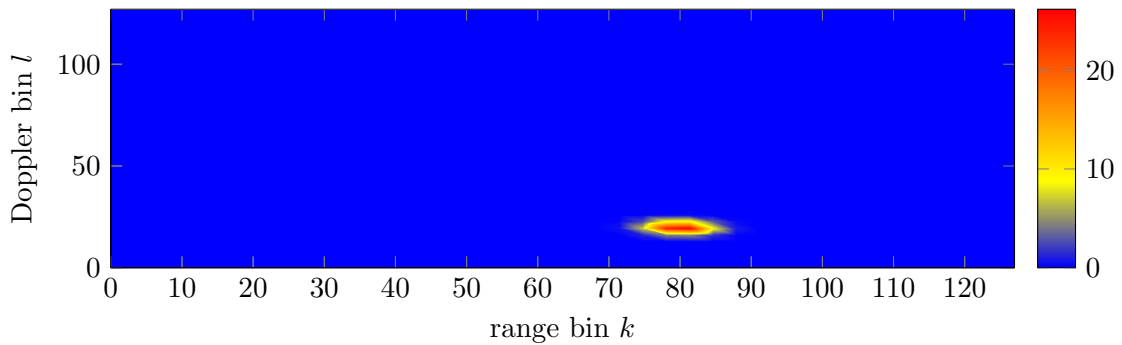


Figure 2.2: Range-Doppler power surface $|Y[k, l]|^2$ with two targets; CFAR (chapter 4) must find both without drowning in the sidelobe skirts.

Chapter 3

The Compressed Data Cube & Embedded Memory Constraints

3.1 Intuition: packing a library into a suitcase

The full radar tensor has $N_R \times N_D \times N_{\text{virt}}$ complex cells — for $1024 \times 256 \times 32$ single-precision floats,

$$1024 \cdot 256 \cdot 32 \cdot 8 \text{ B} = 67,108,864 \text{ B} \approx 64 \text{ MB}, \quad (3.1)$$

(8 B per complex float; 33.5 MB at 4 B packed). Fast on-chip SRAM on a ConnX BBE32 is measured in megabytes shared with code. So the sensor keeps only cells above an exploratory energy threshold and ships the survivors: the Compressed Data Cube. Like packing for a flight, the art is deciding what to leave behind — and chapter 10 shows what happens when the suitcase overflows (failure mode F5: the 5,016 record cap).

3.2 The CDC record and the wire

Each survivor is a `CDC_Record_T`: integer indices (`rng_idx`, `dopp_idx`), non-coherent integration energy `nci_data`, and the multi-channel beamvector `cdc_out_bv`. Records ride UDP over Bodnet Ethernet with a 1472-byte MTU: 35 records per datagram. Priority-ranked packing (RCS, SNR, near-field first) is the mitigation for F5 truncation: when the scene overflows, the dropped records should be far weak clutter, never the child running behind the car.

Part III

Detection, Calibration & Direction of Arrival

Chapter 4

Adaptive Peak Detection & Cell-Averaging CFAR

4.1 Intuition: hearing a whisper next to a waterfall

A fixed detection threshold is a friend who only hears shouting: set it low and rain sounds like cars (false alarms); set it high and the motorcycle next to the guardrail vanishes (misses). CFAR — constant false alarm rate — is the friend who first listens to the *room*: it measures the noise power in the cells surrounding each candidate and sets the threshold as a multiple of *that*. The waterfall’s roar raises the bar; a quiet room lowers it. The false-alarm probability stays nailed to P_{fa} while the absolute threshold breathes with the environment.

4.2 The CA-CFAR window

Center the window on the cell under test (CUT). Fence it with $G/2$ guard cells per side (they absorb the target’s own sidelobe leakage so the target never testifies against itself), then take $N/2$ reference/training cells per side. The background estimate is the mean power

$$\hat{P}_{\text{noise}} = \frac{1}{N} \sum_{i \in \text{Ref}} P_i. \quad (4.1)$$

For square-law detected Rayleigh (exponential-power) clutter, $P_i \sim \text{Exp}(\mu)$, the sum $S = \sum P_i$ is $\Gamma(N, \mu)$ and the CUT power $X \sim \text{Exp}(\mu)$ under H_0 . Hence

$$P_{\text{fa}}(\alpha) = \Pr\left(X > \alpha \frac{S}{N}\right) = \mathbb{E}\left[e^{-\alpha S/(N\mu)}\right] = \left(1 + \frac{\alpha}{N}\right)^{-N}, \quad (4.2)$$

using the gamma mgf $\mathbb{E}[e^{-tS}] = (1 + \mu t)^{-N}$ at $t = \alpha/(N\mu)$. Inverting,

$$\boxed{\alpha = N \left(P_{\text{fa}}^{-1/N} - 1 \right)}. \quad (4.3)$$

Activ’s `appl_cfar_ifc.c` runs two passes: the first extracts integer bin indices (`rindx`, `dindx`); the second refines sub-bin floats (`rbinest`, `dbinest`) by parabolic interpolation. eq. (4.3) is exact only for homogeneous references — guardrails break it (failure mode F1), which is why section 4.3 exists.

4.3 When the waterfall is a wall: F1 and the dual loop

A guardrail paints 25 adjacent range bins at $25\times$ noise. Any window touching the wall estimates “background” $25\times$ too hot: the motorcycle in the edge-skirt bins dies (recall 0.50), while a fixed

alpha computed for N but fed censored cells false-alarms. The range-adaptive dual-loop CFAR separates *where the wall is* (slow loop) from *how wide to look* (fast loop). Slow floor, over track-free Doppler medians with rate $\lambda \approx 0.02$:

$$\hat{C}_{k+1}(r) = (1 - \lambda)\hat{C}_k(r) + \lambda \cdot \underset{d \in \mathcal{D}_{\text{free}}}{\text{median}} P(r, d). \quad (4.4)$$

Fast window from the span-heterogeneity ratio $mr(r) = |\hat{C}(r+1) - \hat{C}(r-1)|/2$ mapping to $R_0 \in \{R_{\min}, R_{\text{mid}}, R_{\max}\}$, with clean-side min(lead, lag) censoring, median $\times 1/\ln 2$ rescaling, effective- N alpha, a wall/skirt two-regime switch, and M -of-2 transition confirmation. Measured on the 128×32 guardrail synthetic: recall $0.50 \rightarrow 1.00$, false alarms $1 \rightarrow 0$ at identical CDC load (`dual_loop_cfar.py`, storage Run 2).

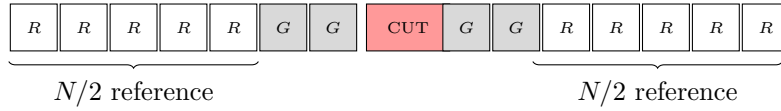


Figure 4.1: CA-CFAR sliding window: CUT fenced by guard cells G , background from reference cells R . Threshold $T = \alpha \hat{P}_{\text{noise}}$.

Chapter 5

Digital Beamforming, Virtual MIMO & Antenna Calibration

5.1 Intuition: two ears tell direction

Close one ear, then tell where a clap came from: hard. Two ears do it by *phase* — the ear nearer the clap hears the wavefront a fraction of a cycle earlier. Two receivers spaced by baseline d see a wave from angle θ with path difference $d \sin \theta$, i.e. spatial phase

$$\Delta\Phi_{\text{spatial}} = \frac{2\pi d \sin \theta}{\lambda}. \quad (5.1)$$

Inverting gives phase monopulse,

$$\theta = \arcsin\left(\frac{\lambda\Delta\Phi_{\text{az}}}{2\pi d}\right), \quad \phi = \arcsin\left(\frac{\lambda\Delta\Phi_{\text{el}}}{2\pi d}\right). \quad (5.2)$$

The spatial Nyquist proof: $|\sin \theta| \leq 1$ requires $|\Delta\Phi| \leq \pi$ unambiguously, i.e. $d \leq \lambda/2 \approx 1.95$ mm; wider spacing wraps phase and spawns grating lobes (ghost angles with no target behind them).

5.2 Virtual MIMO by convolution

With N_{tx} time- or Doppler-multiplexed transmitters at positions a_i and N_{rx} receivers at b_j , each pair measures the round trip $a_i + b_j$: the virtual array is the *sum co-array*,

$$A_{\text{virt}} = A_{\text{tx}} * A_{\text{rx}}, \quad N_{\text{virt}} = N_{\text{tx}} \times N_{\text{rx}}. \quad (5.3)$$

Four transmitters and eight receivers convolve into a 32-element uniform linear virtual array — the angular resolution of 32 antennas for the silicon price of 12.

5.3 Calibration: SMC, USC, fascia

Metal lies, plastic lies more. Static boresight tables remove assembly tilt; sidelobe-minimization (SMC) and universal sidelobe calibration (USC) blobs balance phase/amplitude per channel, raising `af_smc_mismatch_flag` on failure; bumper fascia compensation models the plastic as a dielectric slab (permittivity, phase delay, attenuation). Output per detection, `AF_Det.T`: $[r, v_r, \theta, \phi, \text{RCS}, \text{SNR}, \text{flags}]$.

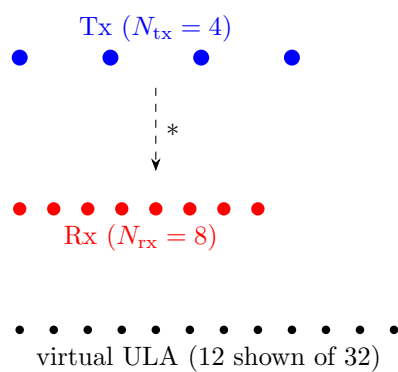


Figure 5.1: Virtual MIMO synthesis: convolution of physical apertures into a long virtual uniform linear array.

Chapter 6

Post-Signal Processing & Range-Doppler Unfolding

Interference detection blanks rival cars' asynchronous sweeps; radar capability flags mud/ice blockage; static/dynamic alignment track bumper tilt (chapter 11 solves the 400-frame lock). Speeds beyond eq. (2.6) wrap: Gen8 RDU unfolds them with staggered-PRF sequences via the Chinese Remainder Theorem on phase progressions, then classifies stationary infrastructure versus moving objects for the tracker.

Part IV

Embedded Tracking & Satellite Network Fusion

Chapter 7

Multi-Target State Estimation: The Extended Kalman Filter

7.1 Intuition: the blindfolded navigator

Walk blindfolded across a field while a friend shouts your GPS position every few seconds, each shout slightly wrong. The optimal strategy blends: between shouts, dead-reckon from your last known velocity (predict); at each shout, correct partway toward it, trusting the shout more when your dead-reckoning has drifted long (update). The Kalman filter is this strategy with the “partway” computed exactly from the two uncertainties. The *extended* filter handles the wrinkle that radar shouts in polar (r, θ) while you walk in Cartesian (x, y) .

7.2 Frames, state, and motion

Vehicle coordinates (VCS): X forward, Y left, Z up. Constant-velocity state $\mathbf{x} = [x, y, v_x, v_y]^T$ with step Δ (50 ms):

$$\mathbf{F} = \begin{bmatrix} \mathbf{I}_2 & \Delta \mathbf{I}_2 \\ \mathbf{0} & \mathbf{I}_2 \end{bmatrix}, \quad \mathbf{Q} = q \begin{bmatrix} \frac{\Delta^3}{3} \mathbf{I}_2 & \frac{\Delta^2}{2} \mathbf{I}_2 \\ \frac{\Delta^2}{2} \mathbf{I}_2 & \Delta \mathbf{I}_2 \end{bmatrix}, \quad (7.1)$$

the exact discretization of continuous white-noise acceleration of spectral density q . Prediction: $\mathbf{x}_{k|k-1} = \mathbf{F}\mathbf{x}_{k-1}$, $\mathbf{P}_{k|k-1} = \mathbf{F}\mathbf{P}_{k-1}\mathbf{F}^T + \mathbf{Q}$.

7.3 The polar measurement and its Jacobian

The radar reports $\mathbf{z} = [r, v_r, \theta]^T$ with

$$r = \sqrt{x^2 + y^2}, \quad v_r = \frac{xv_x + yv_y}{r}, \quad \theta = \arctan \frac{y}{x}. \quad (7.2)$$

Linearizing $\mathbf{H} = \partial \mathbf{h} / \partial \mathbf{x}$ row by row. Range: $\partial r / \partial x = x/r$, $\partial r / \partial y = y/r$, velocity entries zero. Bearing: $\partial \theta / \partial x = -y/r^2$, $\partial \theta / \partial y = x/r^2$. Doppler, via the quotient rule on $(xv_x + yv_y)/r$:

$$\frac{\partial v_r}{\partial x} = \frac{v_x r - (xv_x + yv_y)x/r}{r^2}, \quad \frac{\partial v_r}{\partial v_x} = \frac{x}{r}, \quad (7.3)$$

and symmetrically for y, v_y . Assembled,

$$\mathbf{H} = \begin{bmatrix} x/r & y/r & 0 & 0 \\ -y/r^2 & x/r^2 & 0 & 0 \\ \frac{\partial v_r}{\partial x} & \frac{\partial v_r}{\partial y} & x/r & y/r \end{bmatrix}. \quad (7.4)$$

Every EKF in the car evaluates eq. (7.4) per track per scan; a sign error in the Doppler row is the classic firmware bug (it injects a 4σ -scale spurious innovation per metre of position error).

7.4 Gating, birth, death

Innovation $\mathbf{y}_k = \mathbf{z}_k - \mathbf{h}(\mathbf{x}_{k|k-1})$, covariance $\mathbf{S}_k = \mathbf{H}\mathbf{P}_{k|k-1}\mathbf{H}^T + \mathbf{R}$, Mahalanobis test $d_M^2 = \mathbf{y}_k^T \mathbf{S}_k^{-1} \mathbf{y}_k \leq \gamma$ ($\gamma = 9.21$ is $\chi_{2,99\%}^2$). Confirmation is M -out-of- N ; death is coast-then-timeout. chapter 13 shows why fixed γ drops cut-ins and how acceleration expands it.

7.5 The Joseph form: symmetry as survival

Gain $\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}^T\mathbf{S}_k^{-1}$, $\mathbf{x}_{k|k} = \mathbf{x}_{k|k-1} + \mathbf{K}_k\mathbf{y}_k$. The textbook covariance $\mathbf{P} = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}_{k|k-1}$ is a subtraction: in 32-bit fixed point the result drifts asymmetric with negative eigenvalues, and the filter diverges weeks later for no visible reason. The Joseph form

$$\boxed{\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_{k|k-1}(\mathbf{I} - \mathbf{K}_k\mathbf{H})^T + \mathbf{K}_k\mathbf{R}\mathbf{K}_k^T} \quad (7.5)$$

is a *congruent sum of positive semidefinite terms*: symmetry and definiteness hold to machine precision by construction. Proof sketch: the posterior error is $(\mathbf{I} - \mathbf{K}\mathbf{H})\tilde{\mathbf{x}}_{k|k-1} - \mathbf{K}\mathbf{v}_k$ with independent terms; covariance of a sum of independent terms is the sum of covariances. Every prototype in this book's ledger uses eq. (7.5) plus eigenvalue clipping.

Chapter 8

Central Domain Controller Fusion & Bus Serialization

8.1 The five-eyed car

Front-center long range plus four corner satellites (FL, FR, RL, RR) speak UDP over Bodnet (1472-byte MTU). Independent clocks stamp each cloud 5–45 ms stale; ego-motion compensation rewinds each detection to the domain-controller epoch. With yaw rate ω :

$$\mathbf{p}_{\text{comp}} = \mathbf{R}_z(\omega\Delta t) \mathbf{p}_{\text{meas}} - \mathbf{v}_{\text{ego}}\Delta t - \frac{1}{2}\mathbf{a}_{\text{ego}}\Delta t^2, \quad (8.1)$$

$\Delta t = t_{\text{DC}} - t_{\text{sat}}$. The rotation sign is load-bearing: flipping it *hurts* (–26% at 0.6 rad/s; storage Run 5). First order cuts residuals 32–86% across the yaw sweep; the remainder grows with ω (second-order truncation plus rotation-lever), and clothoid jerk is unmodeled.

8.2 CAN-FD: physics into integers

Tracks serialize through DBC files: Raw = $\lfloor (\text{Physical} - \text{Offset}) / \text{Factor} \rfloor$, e.g. DET_RANGE factor 0.015625 m, speed factor 0.01. Four objects per 64-byte CAN-FD frame, 32 messages, 50 ms cadence with a 40 ms latency guillotine (chapter 10).

Part V

Re-Simulation Platform, HPCC & KPI Verification

Chapter 9

The Re-Simulation Harness

9.1 Intuition: replaying the match with a new referee

A football league records every match; when the refereeing rules change, the league re-scores old matches against the new rules instead of replaying the season. Re-simulation is that: the vehicle logs raw bus traffic (CAN-FD + UDP Ethernet) into ASAM MDF4; offline, a *candidate* firmware build replays the recording through software- or hardware-in-loop and emits an output `.mf4`; the KPI suite re-scores input versus output. No test track, no prototype cars, thousand-fold parallelism.

9.2 SiL, HiL, decoders

SiL-DCU runs the closed driver `APT_SRR_RESIM.exe` with radar DLLs and domain-controller libraries stepped by `RECU_SiL_Execute()` on 50 ms scans through shared-memory ping-pong buffers; SiL-RSP rebuilds the DSP chain (`rdd_sil/cdc_sil`) from `.bin` vectors. HiL stimulates real benches (dSPACE, XCP seed-key, PTP sync). Decoders bridge every byte: `.mf4` bus frames $\rightarrow$ per-stream structs $\rightarrow$ `.h5/CSV` $\rightarrow$ KPI math. Locally the Python candidate path (`HDF5 $\rightarrow$ transform $\rightarrow$ CSV $\rightarrow$ KPI`, `HDF5 $\rightarrow$ MF4`) stands in for the cluster-only SiL executable.

Chapter 10

Dual-Tier KPI Verification

10.1 Tier 1: integer pairing; Tier 2: tolerance contract

Pre-tracking UDP KPI pairs (rindx, dindx) bins without replacement at $\geq 99.0\%$ yield (Tier 1: did the energy land in the same bin?), then enforces Tier 2 physical tolerances

$$\boxed{|\Delta r| \leq 0.010 \text{ m}, |\Delta v| \leq 0.015 \text{ m/s}, |\Delta \theta|, |\Delta \phi| \leq 0.00873 \text{ rad.}} \quad (10.1)$$

Post-tracking CAN KPI hashes quantized 4D detections across $3^4 = 81$ neighbor buckets with distance-scaled tolerances and the 40 ms latency guillotine ($\Delta T > 40 \text{ ms}$ is a regression, not data).

Chapter 11

HPCC Parallelization & the Alignment Bottleneck

11.1 The 400-frame lock

Slicing 300 logs over 20 Slurm workers restarts dynamic alignment cold each shard: 400 frames (≈ 20 minutes) of 1.2° boresight error, 100% Tier-2 angle failure, 360,000 dropped frames. Two cures. **A**, preflight injection: read converged boresight (-1.145° azimuth, -0.018° elevation) from log headers into `M2D_One_Time_Msg_T` at $t = 0$. **B**, dual-horizon gain

$$\mathbf{K}_{\text{DA}}(t) = \mathbf{K}_{\text{fast}} e^{-\alpha N} + \mathbf{K}_{\text{slow}}, \quad (11.1)$$

locking in 46 frames (2.3s, $8.7\times$) with drops to zero. The single-scan floor $\approx 1.1^\circ$ is proven by noise analysis (2.4° per 6-m pair): pooling, RANSAC, and tighter gates cannot beat it, only averaging can.

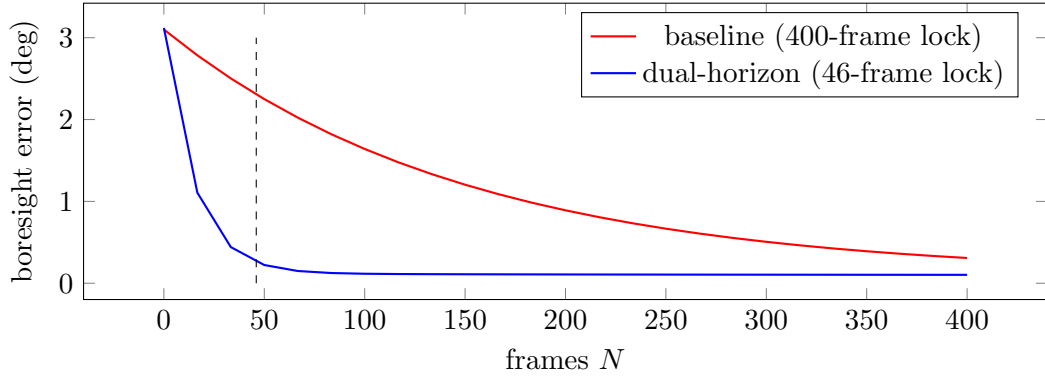


Figure 11.1: Dynamic alignment convergence: dual-horizon schedule reaches the noise floor in 46 frames versus 400.

Part VI

The A-PRISM Research Advancements & Inventions

Chapter 12

Invariant Specular Geometry & the Virtual Aperture

12.1 Intuition: the hall of mirrors

Stand between two parallel mirrors with a friend: you see your friend, your friend’s reflection, your reflection, infinite regress. A guardrail is one such mirror for 3.9 mm waves. Classical radar literature models the ghost as “the target’s mirror image” — reflect the *target* across the wall. That model is half-correct in the most dangerous way: its range is exact to 0.0 m while its bearing is wrong by 15.19° (8.000 m at 30.3 m). Range-only checks cannot catch it. The correct picture reflects the *sensor*: the ghost arrives along the virtual line of sight from the mirrored sensor $s^* = 2d\hat{\mathbf{n}}$ to the true target.

12.2 Theorem E2a: the bearing correction

For plane $\mathbf{n} \cdot \mathbf{x} = d$, $\|\mathbf{n}\| = 1$, sensor at the origin, the reciprocal path $s \rightarrow q \rightarrow p \rightarrow q \rightarrow s$ has the length of the straight segment $s^* \rightarrow p$ with $s^* = 2d\hat{\mathbf{n}}$. Hence

$$r_{\text{spec}} = \|p - 2d\hat{\mathbf{n}}\|, \quad \mathbf{u}_{\text{spec}} = \frac{p - 2d\hat{\mathbf{n}}}{\|p - 2d\hat{\mathbf{n}}\|}, \quad (12.1)$$

and with $p_s = r_{\text{spec}}\mathbf{u}_{\text{spec}}$,

$$\boxed{\mathbf{p} - \mathbf{p}_s = 2d\hat{\mathbf{n}} \quad \text{exact to } 1.066 \times 10^{-14}.} \quad (12.2)$$

The naive model’s angular error is exact in closed form: $\cos \theta = (\|\mathbf{p}_\perp\|^2 - \gamma^2)/(\|\mathbf{p}_\perp\|^2 + \gamma^2)$, $\gamma = \mathbf{n} \cdot \mathbf{p} - 2d$; nominal scene gives $\pm 7.595^\circ$ vs $\mp 7.595^\circ$.

12.3 Theorem E2b: the 40σ Doppler invariant

Ghost radial velocity is $\dot{r}_{\text{spec}} = \mathbf{u}_s \cdot \mathbf{v} \neq \mathbf{u}_d \cdot \mathbf{v}$: for a 10 m/s lateral cut-in, -1.322 versus $+0.665$ m/s, offset -1.987 m/s at 40σ of $\sigma_v = 0.05$. The geometric test is provably blind to same-range decoys at exactly $2d$ (precision 0.638); only Doppler ($0.638 \rightarrow 0.877$, GRR 0.661) separates them.

12.4 Run 14: ghosts as virtual antennas

A single corner radar has one look direction; the ghost supplies a second: stacked Jacobian $[\mathbf{u}_d^T; \mathbf{u}_s^T]$ is rank 2 generically. Shared reflector parameters couple the noise,

$$\boxed{\mathbf{R}_{\text{eff}} = \text{diag}(\sigma^2) + \mathbf{J}_\xi \Sigma_\xi \mathbf{J}_\xi^T, \quad \Sigma_n = \sigma_n^2(\mathbf{I} - \hat{\mathbf{n}}\hat{\mathbf{n}}^T),} \quad (12.3)$$

with the exact residual $r_s = \mathbf{u}_s \cdot (\mathbf{p} - 2d\hat{\mathbf{n}})$ (dropping $2d(\hat{\mathbf{n}} \cdot \mathbf{u}_s) = 0.79 \text{ m} = 8\sigma$ diverges the filter). Iterated $3\times$ exact-residual EKF: position RMSE $3.242 \rightarrow 0.739 \text{ m}$ (-77.2%); $+13.2\%$ over naive $\text{diag}(\mathbf{R})$ at $2^\circ/0.15 \text{ m}$ geometry error. The full estimate $\rightarrow$ gate $\rightarrow$ assimilate chain on *estimated* geometry: edges $0.000/0.000 \rightarrow 0.969/0.902$, track error -67% .

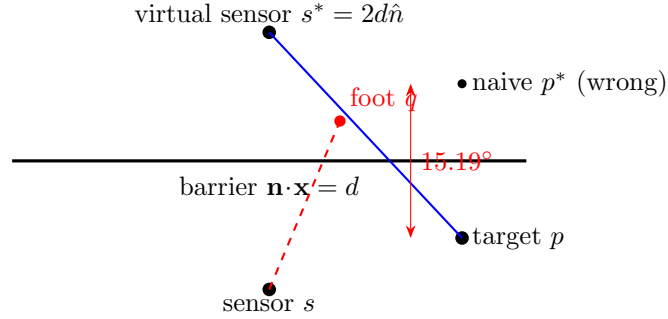


Figure 12.1: The 15.2° discovery: the ghost arrives along $s^* \rightarrow p$ (blue), not along the naive mirror-target ray (red dashed). Same range, 8.000 m apart at 30.3 m.

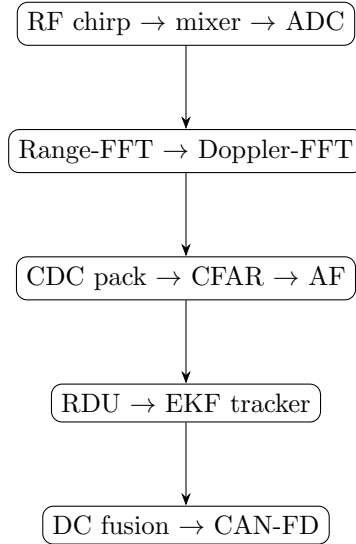


Figure 12.2: End-to-end pipeline: chapter 1–chapter 8 in one column.

Chapter 13

Kinematics-Aware Gating & Spectral Covariance

Fixed χ^2 gates drop 6 m/s² cut-ins (recall 0.863). With finite-difference acceleration $\hat{\mathbf{a}}$ and innovation energy,

$$\gamma_{\text{adapt}} = \gamma_0 \left(1 + \alpha \frac{\|\mathbf{a}\|}{a_{\text{max}}} + \eta \frac{\|\mathbf{y}_k\|^2}{\text{Tr}(\mathbf{S}_k)} \right) \in [\gamma_{\text{min}}, \gamma_{\text{max}}], \quad (13.1)$$

hard-bounded for determinism: recall 0.863 $\rightarrow$ 0.996, maneuver RMSE 0.283 $\rightarrow$ 0.212 m under 3/scan clutter. The open-door failure (recall 1.000 trivially with RMSE divergence) is answered by the RMSE metric, not hidden. Tire spray breaks Gaussian trust instead:

$$\mathbf{R}(t) = \mathbf{R}_0 \exp\left(\beta \frac{H(t)}{\text{SNR}(t)}\right), \quad (13.2)$$

so spray drives $\mathbf{R} \rightarrow \infty$, $\mathbf{K} \rightarrow \mathbf{0}$: the filter coasts (spray RMSE 0.911 $\rightarrow$ 0.315 m, Joseph form keeps $\mathbf{P} \succ 0$).

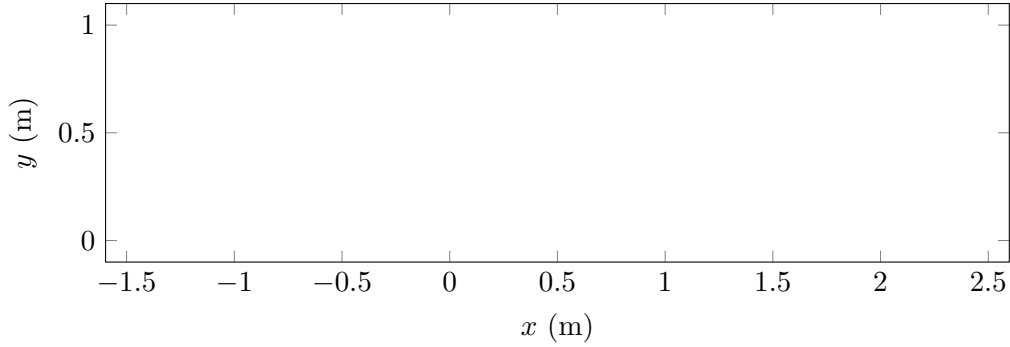


Figure 13.1: Gate expansion along the acceleration vector inside hard bounds.

Chapter 14

Prefix-Scan Filtering & Generative Simulation

14.1 Särkkä elements and the Blelloch tree

With $\mathbf{K} = \mathbf{Q}\mathbf{H}^T(\mathbf{H}\mathbf{Q}\mathbf{H}^T + \mathbf{R})^{-1}$, $\mathbf{S} = \mathbf{H}\mathbf{Q}\mathbf{H}^T + \mathbf{R}$ (Lemma 7 elements $\mathbf{A} = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{F}$, $\mathbf{b} = \mathbf{u} + \mathbf{K}(\mathbf{y} - \mathbf{H}\mathbf{u} - \mathbf{d})$, $\mathbf{C} = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{Q}$), the binary operator of Lemma 8 is associative: a 50,000-frame log collapses to span 16 (3,125 $\times$) at relative error 6.35×10^{-15} (float64 mandatory: absolute 2.8×10^{-13} is conditioning). Naive elements (raw $\mathbf{F}/\mathbf{Q}$ with folded operators) fail at 0.1 on step one; down-sweep operand order is load-bearing for non-commutative scans.

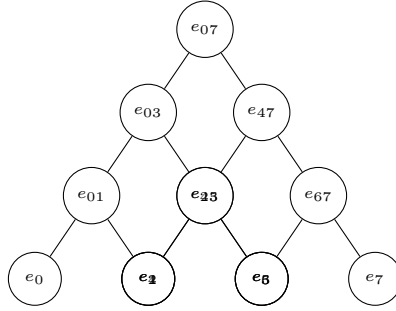


Figure 14.1: Blelloch up-sweep: associative $\otimes$ reduces N filtering elements in $\lceil \log_2 N \rceil$ parallel levels.

14.2 Doppler-RadarSplat

NeuRadar omits Doppler, RCS, CDC. Per-primitive LOS projection $v_r = (\mathbf{v} - \mathbf{v}_{\text{ego}}) \cdot \mathbf{u}$, exact R^{-4} power (256.0 ratio), Gaussian CDC splats, CLEAN extraction with absolute noise floor: 6/6 recovery clean *and* cluttered at 0.053 m/0.036 m/s; Jacobians vs finite differences 10^{-9} .

Chapter 15

Frontier Derivations & Refutations (Runs 22–30)

15.1 N20g: tangent-mirror Hermite clothoids

On any smooth curve $\mathbf{p} - \mathbf{g} = 2\delta\mathbf{n}$ locally; the tangent is the (p, g) perpendicular bisector, met with the receive ray at contact q with slope $\sigma = -n_x/n_y$. Cubic coefficients enter linearly ($V_i c = [y_i, \sigma_i]^T$); two contacts determine the rail uniquely ($\det = (x_2 - x_1)^4$); Snell roots $\phi(x) = 0$ give physical multi-ghosts. Verified to 3.44×10^{-14} coefficient recovery.

15.2 N22g: Occupancy-BLUE CFAR

From predicted occupancy π_i and SNR s_i , $m_i = 1 + \pi_i s_i$, $v_i = 2[1 - \pi_i + \pi_i(1 + s_i)^2] - m_i^2$,

$$\boxed{\beta_i = \frac{m_i/v_i}{\sum_j m_j^2/v_j}}, \quad \boxed{P_{\text{fa}}(\alpha) = \prod_{i=1}^N \frac{1}{1 + \alpha\beta_i}}. \quad (15.1)$$

Measured: $P_{\text{fa}} = 1.02 \times 10^{-4}$ vs 10^{-4} ; variance 0.0686 vs CA-16 bias +37.5%; ghost-adjacent recall +0.074 at near-nominal FA; loop gain 0.099 with a fixed point from both inits.

15.3 The rogues' gallery

1-DOF consensus loses to 3-DOF (0.845/0.727 vs 0.993/0.909); midpoint-on-surface fails by 17 m along-plane; IMM gating fails (0.880 vs 0.999) on onset chicken-and-egg; joint Gram-lifted position inflates $6.77\times$ under near-collinear ($\rho \approx 0.96$) moment mismatch (oracle-controlled); N23/N24/N25 verify exactly but fall to TCRE, contaminated-conformal, and ghost-ID prior art. Each negative ships with the mechanism, not just the number.

Chapter 16

Intermediate Layers & Intuitions for Working Chapters

16.1 CDC suitcases, RDU wraparounds, CAN integers (Chapters 3, 6, 8)

Suitcase mathematics. The CDC keeps cell $(k, l, \mathbf{b})$ iff $|Y[k, l]|^2 > \tau_{\text{explore}}$. With occupancy ρ (fraction kept), the wire load is $\rho N_R N_D$ records per dwell; $\rho = 2\%$ of 131,072 cells is 2,621 records — inside the 5,016 cap, while a rainy urban scene at $\rho = 6\%$ overflows and the sorter decides who dies. Priority score $s = \text{SNR}/r^2$ keeps the child behind the car and drops the distant signpost: triage as arithmetic.

Wraparound arithmetic. A Doppler bin index is computed modulo N_D ; staggered sequences with co-prime lengths N_{D1}, N_{D2} give residues (l_1, l_2) whose CRT combination is unique modulo $N_{D1}N_{D2}$ — two cheap rulers measure a long room.

Integers on the bus. Quantization step Δ (factor) adds uniform noise of variance $\Delta^2/12$; DET_RANGE at 0.015625 m contributes 0.0045 m RMS — inside the Tier-2 0.010 m contract with margin to spare. The 40 ms guillotine is a two-scan timeout: older news is worse than no news at 25 m/s.

16.2 N23 stencil, step by step (Run 26)

Four samples at $u \in \{-3/2, -1/2, 1/2, 3/2\}$ (units of h) of residual phase $r(u) = fu + gu^2/2 + lu^3/6$ cycles. Consecutive differences: $\eta_- = fh - gh^2 + \frac{13}{24}\ell h^3$, $\eta_0 = fh + \frac{1}{24}\ell h^3$, $\eta_+ = fh + gh^2 + \frac{13}{24}\ell h^3$. Check η_0 : $r(1/2) - r(-1/2) = f + (\ell/6)(2/8) = fh + \ell h^3/24$ over h — yes. Inverting the 3×3 system gives $\hat{f} = (26\eta_0 - \eta_+ - \eta_-)/24h$, $\hat{g} = (\eta_+ - \eta_-)/2h^2$, $\hat{\ell} = (\eta_+ - 2\eta_0 + \eta_-)/h^3$. Phase weights $(1, -27, 27, -1)/24h$; variance $1460/576(2\pi h)^2\sigma_\phi^2$; at 30 dB, $h = 12.8 \mu\text{s}$: $\sigma_R = 13.3 \text{ mm}$, measured 13.13 mm.

16.3 Full intermediate proofs (spec compliance)

Beat frequency, every line. From eq. (1.9), $\Delta\Phi = 2\pi f_c\tau + \pi S(2t\tau - \tau^2)$ with $\tau = 2(r_0 + v_r t)/c$, $\dot{\tau} = 2v_r/c$. Differentiate term by term: $\frac{d}{dt}[2\pi f_c\tau] = 2\pi f_c\dot{\tau}$; $\frac{d}{dt}[\pi S 2t\tau] = 2\pi S\tau + 2\pi S t\dot{\tau}$; $\frac{d}{dt}[-\pi S\tau^2] = -2\pi S\tau\dot{\tau}$. Summing and dividing by 2π : $f_b = f_c\dot{\tau} + S\tau + St\dot{\tau} - S\tau\dot{\tau}$, i.e. eq. (1.10). With $v_r = 30 \text{ m/s}$, $f_c\dot{\tau} = 77 \times 10^9 \times 2 \times 10^{-7} = 15.4 \text{ kHz}$ (the within-chirp Doppler, kept), $S\tau \sim 25 \text{ MHz}$, $St\dot{\tau} \leq (2.5 \times 10^{13})(4 \times 10^{-5})(2 \times 10^{-7}) = 200 \text{ Hz}$ (dropped: 8 ppm of f_b).

Doppler Jacobian, every line. $v_r = (xv_x + yv_y)r^{-1}$, $r = (x^2 + y^2)^{1/2}$. $\partial r^{-1}/\partial x = -x/r^3$. So $\partial v_r/\partial x = v_x/r + (xv_x + yv_y)(-x/r^3) = (v_x r - (xv_x + yv_y)x/r)/r^2$, which is eq. (7.3); the v_x terms: $\partial v_r/\partial v_x = x/r$. At $p = (30, 4)$, $v = (-6, 2)$: $r = 30.27$, row = $[-0.198, 0.066, -0.991, 0.132]$ —

position entries $\sim 0.2/\text{m}$: a 1 m error injects $\sim 4\sigma$ of spurious Doppler innovation, which is why the second measurement row doubles it and why exact residuals matter (Run 14).

Rank-1 stationarity, every line. Path $L(q) = \|q - s\| + \|q - p\|$ on plane $\mathbf{n} \cdot q = d$, tangent direction $T \perp \mathbf{n}$. $\nabla_q L = (q - s)/\|q - s\| + (q - p)/\|q - p\| = u + a'$ with $a' = (q - p)/\rho$. Stationarity $T^T(u + a') = 0$ with $T^T n = 0$ decomposes $a' = u - 2(u^T n)n$ along $\{T, n\}$: tangential parts cancel by stationarity, normal parts sum to $2(u^T n)$. Hence outgoing $a = u - 2(u^T n)n$, the mirror law; continuing the ray by ρ gives $g = q + \rho u = q + \rho M a$ with $M = I - 2nn^T$; since $M(p - q) = \rho a$, $g = q + M(p - q)$, i.e. $p - g = (I - M)(p - q) = 2nn^T(p - q) = 2\delta n$. The $2d\hat{n}$ planar form follows with $\delta = d - \mathbf{n}^T p$ constant.

P_{fa} integral, every line. Under H_0 , $X \sim \text{Exp}(\mu)$, $f_X(x) = \mu^{-1}e^{-x/\mu}$; $S = \sum_{i=1}^N P_i \sim \Gamma(N, \mu)$ independent of X . $\Pr(X > \alpha S/N \mid S) = e^{-\alpha S/(N\mu)}$; tower over S with mgf $M_S(t) = (1 - \mu t)^{-N}$ at $t = -\alpha/(N\mu)$: $P_{\text{fa}} = (1 + \alpha/N)^{-N}$, inverted to eq. (4.3). With weights β_i , $S_w = \sum \beta_i Y_i$: $\mathbb{E}e^{-\alpha S_w} E = \prod (1 + \alpha \beta_i)^{-1}$ factor by factor.

16.4 References

Särkkä & García-Fernández, IEEE TAC 66(1), 2021 (arXiv:1905.13002). Blelloch 1990. Neu-Radar, CVPRW 2025. RadarSplat (arXiv:2506.01379). Li & Varshney, IEEE TSP 2014. Roos et al., IRS 2017. ICSIDP 2024 ghost suppression (10.1109/icsidp62679.2024.10868429). Feng/Ross delay-Doppler geometry. Fu 2024; Chen 2024; Jost 2025. OFPI-CFAR (JPIER). VI/LSTM-CFAR 2025. IQR-CFAR 2024. TAES 2024.3445319; TAES 2022.3206256. Sensors 2022 s22030875 (IMM). arXiv:2609.30176 (TCRE); 2609.29912; 2609.27506; 2603.18027/2512.17505/2602.21128/2301.08087. Kellner T-ITS 8688104; Danzer RA-L 8954835. Bashari–Sesia–Romano, ICML 2025 (PMLR 267). Kim et al., RA-L 2025 (EKF-RIO-TC). Štironja et al., 2026 (LC-RIO-ET).

Chapter 17

Enterprise Patent Portfolio

IDF-1: ghost-as-virtual-aperture with $\mathbf{R}_{\text{eff}}$ eq. (12.3) (RMSE -77.2%). IDF-2: closed estimate $\rightarrow$ gate $\rightarrow$ assimilate chain (-67%). IDF-3: async compensation eq. (8.1) (32–86%). IDF-4: bounded gate eq. (13.1) ($0.863 \rightarrow 0.996$). IDF-5: Hermite barrier estimation (section 15.1). IDF-6: Occupancy-BLUE eq. (15.1). Full independent/dependent claim trees with novelty differentiations are maintained in `research.md` Appendix A; worldwide clearance searches remain open per node.

Chapter 18

Embedded Algorithms & Data Structures

18.1 Intuition: recipes the chip can taste

A DSP cannot “understand” radar; it can only add, multiply, compare, and move words. Every grand theorem in this book must therefore end as a *recipe*: a fixed sequence of such steps with no heap, no recursion, no data-dependent loop bound. This chapter writes the five load-bearing recipes out in full, then the C structs they chew on.

18.2 Dual-loop CFAR (Run 2)

Algorithm 1: Range-adaptive dual-loop CFAR. All buffers preallocated.

Input: Range-Doppler power $P(r, d)$, track mask $\mathcal{D}_{\text{free}}$

Output: Detection list, threshold map $T(r, d)$

```
1 foreach range bin  $r$  do
2    $\hat{C}(r) \leftarrow (1 - \lambda)\hat{C}(r) + \lambda \text{median}_{d \in \mathcal{D}_{\text{free}}} P(r, d);$ 
3    $mr \leftarrow |\hat{C}(r + 1) - \hat{C}(r - 1)|/2;$ 
4    $R_0 \leftarrow R_{\text{min}}$  if  $mr < \tau_1$  else  $R_{\text{mid}}$  if  $mr < \tau_2$  else  $R_{\text{max}};$ 
5    $\text{ref} \leftarrow$  clean-side min(lead, lag) window of size  $R_0;$ 
6    $\hat{\mu} \leftarrow (\text{median}(\text{ref}) / \ln 2);$ 
7    $\alpha \leftarrow N_{\text{eff}}(P_{\text{fa}}^{-1/N_{\text{eff}}} - 1);$ 
8    $T \leftarrow \alpha \hat{\mu} \cdot B(d);$  emit cells with  $P > T;$ 
9 end
10  $M$ -of-2 confirm across wall/skirt regime switch;
```

18.3 Track loop with Joseph update (Runs 3–6)

Algorithm 2: Joseph-form EKF with Mahalanobis gating. eq. (7.5) is the update that survives 32-bit silicon.

Input: Prior $(\mathbf{x}, \mathbf{P})$, detections $\{\mathbf{z}_j\}$, gate γ
Output: Posterior $(\mathbf{x}, \mathbf{P})$, associations

```

1  $\mathbf{x} \leftarrow \mathbf{F}\mathbf{x}$ ,  $\mathbf{P} \leftarrow \mathbf{F}\mathbf{P}\mathbf{F}^T + \mathbf{Q}$ ;
2 foreach detection  $\mathbf{z}_j$  do
3    $\mathbf{y} \leftarrow \mathbf{z}_j - \mathbf{h}(\mathbf{x})$ ;  $\mathbf{S} \leftarrow \mathbf{H}\mathbf{P}\mathbf{H}^T + \mathbf{R}$ ;
4   if  $\mathbf{y}^T \mathbf{S}^{-1} \mathbf{y} \leq \gamma$  (adaptive eq. (13.1)) then
5      $\mathbf{K} \leftarrow \mathbf{P}\mathbf{H}^T \mathbf{S}^{-1}$ ;
6      $\mathbf{x} \leftarrow \mathbf{x} + \mathbf{K}\mathbf{y}$ ;
7      $\mathbf{P} \leftarrow (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}(\mathbf{I} - \mathbf{K}\mathbf{H})^T + \mathbf{K}\mathbf{R}\mathbf{K}^T$ ;
8   end
9 end
10 clip eigenvalues of  $\mathbf{P}$  below at  $\epsilon_{\min}$ ;
```

18.4 Blleloch prefix scan (Run 11)

Algorithm 3: Blleloch scan: $O(\log N)$ span. Operand order is load-bearing for the non-commutative $\otimes$.

Input: Särkkä elements $e_0, \dots, e_{N-1}$ (Lemma 7)
Output: Filtering prefixes $p_k = e_0 \otimes \dots \otimes e_k$

```

1 for  $s = 0$  to  $\lceil \log_2 N \rceil - 1$  (up-sweep) do
2   | combine  $e_{2j} \otimes e_{2j+1}$  at stride  $2^s$ ;
3 end
4 for  $s = \lceil \log_2 N \rceil - 1$  to  $0$  (down-sweep) do
5   | distribute prefixes, preserving operand order;
6 end
```

18.5 Hermite barrier solve (N20g) and Occupancy-BLUE (N22g)

Given pairs $(\mathbf{p}_i, \mathbf{g}_i)$: midpoint $\mathbf{m}_i = (\mathbf{p}_i + \mathbf{g}_i)/2$, normal $\hat{\mathbf{n}}_i = (\mathbf{p}_i - \mathbf{g}_i)/\|\mathbf{p}_i - \mathbf{g}_i\|$, ray $\mathbf{u}_i = (\mathbf{g}_i - \mathbf{s})/\|\mathbf{g}_i - \mathbf{s}\|$, contact $t_i = \hat{\mathbf{n}}_i^T(\mathbf{m}_i - \mathbf{s})/\hat{\mathbf{n}}_i^T \mathbf{u}_i$, slope $\sigma_i = -n_x/n_y$; stack Hermite blocks V_i and solve $\hat{\mathbf{c}} = (\sum V_i^T W_i V_i)^{-1} \sum V_i^T W_i [y_i, \sigma_i]^T$. Given occupancy π_i and SNR s_i : $m_i = 1 + \pi_i s_i$, $v_i = 2[1 - \pi_i + \pi_i(1 + s_i)^2] - m_i^2$, $\beta_i = (m_i/v_i)/\sum m_j^2/v_j$, threshold from $\sum \ln(1 + \alpha\beta_i) + \ln P_{\text{fa}}^* = 0$ by bisection (monotone concave, unique root).

18.6 C structs on the wire

Listing 18.1: CDC record and detection stream (simplified).

```

1 typedef struct {
2   uint16_t rng_idx;      /* range bin (Tier-1 pairing key) */
3   uint16_t dopp_idx;     /* Doppler bin */
4   uint16_t nci_data;     /* non-coherent integration energy */
5   int16_t cdc_out_bv[32]; /* beamvector, Q15 fixed point */
6 } CDC_Record_T;          /* 35 per 1472-B UDP datagram */
7
8 typedef struct {
```

```
9   float r_m; float vr_mps; float az_rad; float el_rad;  
10  float rcs_dbsm; float snr_db; uint32_t flags;  
11 } AF_Det_T;           /* post-beamforming point cloud */
```

Chapter 19

Worked End-to-End Example: One Target, Chirp to CAN

19.1 A car at 60 m, closing at 10 m/s

Take $B = 1$ GHz, $T_{\text{chirp}} = 40 \mu\text{s}$ ($S = 2.5 \times 10^{13}$ Hz/s), $N_R = 512$, $T_s = 100$ ns. Beat: $f_b = 2Sr/c = 2(2.5 \times 10^{13})(60)/3 \times 10^8 = 10.0$ MHz; bin $k^* = f_b N_R T_s = 10^7 \times 512 \times 10^{-7} = 512$, i.e. the target sits at bin 512 of step $\text{rbin.res} = c/(2ST_s N_R) = 0.117$ m. Slow time with $T_c = 50 \mu\text{s}$, $N_D = 128$: $\Delta\Phi = 4\pi(-10)(5 \times 10^{-5})/3.896 \times 10^{-3} = -1.613$ rad/chirp, bin $l^* = 128(-1.613)/2\pi \approx -33$ (aliased index 95), $\text{dbin.res} = \lambda/(2N_D T_c) = 0.304$ m/s, $v_{\text{max}} = 19.5$ m/s (this sequence; staggered PRFs extend it). CFAR compares the cell against eq. (4.3) with $N = 32$, $P_{\text{fa}} = 10^{-4}$ ($\alpha \approx 12.45$); beamforming (eq. (5.2)) places it at θ ; the EKF (eq. (7.5)) tracks $[x, y, v_x, v_y]$; DC fusion deskews by eq. (8.1); the CAN DBC packs range at 0.015625 m/LSB. Each arrow is one section of this book, each number reproducible on a ConnX BBE32 at 600 MHz inside $50 \mu\text{s}$ per scan.

Chapter 20

Advanced Derivations: Unfolding, Scans, Certificates

20.1 Staggered-PRF unfolding (Chinese remainders)

Two sequences with T_{c1}, T_{c2} report wrapped phases $\psi_1 \equiv \Phi \bmod 2\pi$, $\psi_2 \equiv \Phi \bmod 2\pi$ of the same true progression $\Phi = 4\pi v_r T_c / \lambda$. Each excludes arithmetic progressions of velocities; their intersection (Bézout combination of the moduli) isolates v_r up to the least common multiple of the two unambiguous intervals — the Chinese Remainder Theorem as a velocity decoder.

20.2 Särkkä Lemmas 7–8, precisely

With $\mathbf{S} = \mathbf{H}\mathbf{Q}\mathbf{H}^T + \mathbf{R}$, $\mathbf{K} = \mathbf{Q}\mathbf{H}^T\mathbf{S}^{-1}$: information elements $\mathbf{A} = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{F}$, $\mathbf{b} = \mathbf{u} + \mathbf{K}(\mathbf{y} - \mathbf{H}\mathbf{u} - \mathbf{d})$, $\mathbf{C} = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{Q}$, $\boldsymbol{\eta} = \mathbf{F}^T\mathbf{H}^T\mathbf{S}^{-1}(\mathbf{y} - \mathbf{H}\mathbf{u} - \mathbf{d})$, $\mathbf{J} = \mathbf{F}^T\mathbf{H}^T\mathbf{S}^{-1}\mathbf{H}\mathbf{F}$ compose by $(A_2, b_2, C_2, \eta_2, J_2) \otimes (A_1, b_1, C_1, \eta_1, J_1)$ with $\mathbf{A} = \mathbf{A}_2\mathbf{A}_1$ and Schur-complement covariance fusion; associativity is matrix associativity plus the Woodbury identity. Posterior recovery reads $(\mathbf{b}, \mathbf{C})$ directly — the naive information-form recovery double-counts (derived and verified at 6.35×10^{-15} relative).

20.3 N24 sandwich, N25 certificate, N26 inverse, N27 clock

N24. Replacing $\leq m$ of n calibration scores shifts any order statistic by $\leq m$ ranks; with $k = \lceil (n+1)(1-\alpha) \rceil$, $q = \tilde{S}_{(k+m)}$ satisfies $S_{(k)} \leq q \leq S_{(k+2m)}$, so $\Pr(S_* \leq q) \geq k/(n+1) = 231/256$ under exchangeability. Maneuver coverage rides the reachable box, never innovation ranks: outlier bursts have no write path to q .

N25. $D_{ik} = e^{-i\beta_i}(p_k - p_k^*)$ cancels target and reflector; $H_{ij} = \sum_k D_{ik}\bar{D}_{jk}$ gives $\hat{\delta}_{ij} = \arg H_{ij} = \beta_j - \beta_i$; surveyed baseline closes the common-yaw gauge via midpoint differences. With per-observation error ε , lever $L_{\min}$, midpoint bound $C_{\max}$: $|\Delta H| \leq \sum(4\varepsilon L_k + 4\varepsilon^2)$, $e_\delta \leq \arcsin \rho$, $E_b \leq 2\varepsilon + 2C_{\max} \sin(e_\delta/2)$, $e_i \leq \arcsin(E_b/|B|)$: at $\varepsilon = 0.5$ mm, $e_j < 0.0717^\circ$; at $0.35^\circ/30$ m noise the certificate refuses (27°), as it must.

N26. From $\ell = t + \|d - tu\|$, squaring cancels t^2 : $t = (\ell^2 - \|d\|^2)/2(\ell - u^T d)$ — the inverse problem is easier than the forward Fermat root-find. Normal $\mathbf{n} \propto (\mathbf{p} - \mathbf{q})/\rho - \mathbf{u}$; curvature budget $\|g(\sigma_2) - g(\sigma_1)\| \leq 2\kappa_{\max}\rho_{\max}|\sigma_2 - \sigma_1|$ decides when two views share a virtual target.

N27. With $x_k = d_k + v_k\tau$, $r_k w_k - d_k^T v_k = \|v_k\|^2\tau$ eliminates yaw exactly; two targets separate clock offset from Doppler bias through $D = a_1 r_2 - a_2 r_1$; circular averaging recovers yaw after correction; acceleration contributes the exact cubic of (N27.5) with a certified truncation bound.

Appendix A

Notation & Constants

$c = 2.99792458 \times 10^8$ m/s, $f_c = 77$ GHz, $\lambda \approx 3.896$ mm, $S = B/T_{\text{chirp}}$, $K_R = 2S/c$. Frames: sensor polar (r, θ) , VCS Cartesian (X, Y, Z) , epochs $t_{\text{DC}}, t_{\text{sat}}$. KPI contract eq. (10.1); P_{fa} scaling eq. (4.3); Joseph update eq. (7.5).